

AP Statistics MCQ Practice Exam 12

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PART A, QUESTIONS

APSTAT-MCQ12-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The five-number summary of a variable X is minimum 4, $Q1 = 7$, median 10, $Q3 = 14$, and maximum 20. A new variable is defined by $Y = 60 - 2X$. What is the five-number summary of Y ?

- A) Minimum 52, $Q1 = 46$, median 40, $Q3 = 32$, maximum 20
- B) Minimum 20, $Q1 = 32$, median 40, $Q3 = 46$, maximum 52
- C) Minimum 20, $Q1 = 46$, median 40, $Q3 = 32$, maximum 52
- D) Minimum 4, $Q1 = 7$, median 10, $Q3 = 14$, maximum 20

APSTAT-MCQ12-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A 95% confidence interval for $p_{\text{text}} - p_{\text{email}}$, the difference in reminder-response proportions, is $(-0.12, -0.04)$. Which interpretation is supported?

- A) Text reminders have a response proportion between 4 and 12 percentage points higher than email reminders.
- B) Exactly 95% of people respond differently under the two reminder methods.
- C) The two population response proportions are equal because both endpoints are negative.
- D) The text-reminder population proportion is estimated to be 4 to 12 percentage points lower than the email-reminder population proportion.

APSTAT-MCQ12-003

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A box contains 12 tags: 3 amber and 9 non-amber. Two tags are selected without replacement. Given that at least one selected tag is amber, what is the probability that both are amber?

- A) $1/10$
- B) $1/6$
- C) $3/11$
- D) $5/11$

APSTAT-MCQ12-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Each audio file is processed by a manual method and an automated method. Let d = automated processing time – manual processing time for a file. Researchers want to test whether the automated method reduces the population mean processing time. Which hypotheses are appropriate?

- A) $H_0: \mu_d = 0$; $H_a: \mu_d > 0$
- B) $H_0: \bar{x}_d = 0$; $H_a: \bar{x}_d < 0$
- C) $H_0: \mu_d = 0$; $H_a: \mu_d < 0$
- D) $H_0: \mu_{\text{automated}} = \mu_{\text{manual}}$; $H_a: \mu_{\text{automated}} \neq \mu_{\text{manual}}$, treating the samples as independent

APSTAT-MCQ12-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For a dataset relating x to y , the least-squares line is $\hat{y} = 18 + 1.6x$. The sample mean of x is $\bar{x} = 5$. What is the sample mean of y ?

- A) 23
- B) 26
- C) 28.8
- D) It cannot be determined from the fitted line and $\bar{x}$.

APSTAT-MCQ12-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A city wants to estimate the proportion of rented apartments needing a safety repair. Its address file lists each building once, whether the building contains 1 apartment or 80 apartments. Which sampling plan best aligns selection with the population of apartments?

- A) Construct a list of individual apartment units in the study area and randomly select units from that list.
- B) Randomly select buildings from the address file and inspect exactly one apartment in each without weighting.
- C) Inspect every apartment in the first building whose owner agrees to participate.
- D) Ask property managers to nominate apartments they consider typical.

APSTAT-MCQ12-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

For a two-proportion z-interval, sample A has 44 successes and 16 failures, while sample B has 23 successes and 7 failures. Which assessment is correct?

- A) All large-count conditions are met because the combined number of observations exceeds 30.
- B) The procedure is invalid only because the two sample sizes are unequal.
- C) The procedure is valid because each sample has at least 10 successes.
- D) The usual large-count condition is not met because sample B has only 7 failures.

APSTAT-MCQ12-008

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For repeated simple random samples of a fixed size, the sampling distribution of $\hat{p}$ has mean 0.38 and standard deviation 0.0485. What does the standard deviation describe?

- A) In 4.85% of samples, $\hat{p}$ will equal 0.38.

- B) The population proportion changes by 0.0485 from sample to sample.
- C) Values of $\hat{p}$ typically vary by about 0.0485 from the population proportion 0.38 across repeated samples.
- D) Every sample proportion lies between 0.3315 and 0.4285.

APSTAT-MCQ12-009

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A sample of 10 measurements has $\Sigma x = 70$ and $\Sigma x^2 = 530$. What is the sample standard deviation?

- A) 2.00
- B) 2.05
- C) Approximately 2.11
- D) Approximately 4.44

APSTAT-MCQ12-010

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A one-sample t statistic is 1.30 for a sample of $n = 25$. A second random sample from the same population has the same sample mean and sample standard deviation but $n = 100$. For the same null value, what is the second t statistic?

- A) 0.65
- B) 2.60
- C) 5.20
- D) 1.30

APSTAT-MCQ12-011

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

A computer summary for two quantitative variables reports a least-squares slope of -1.8 and a correlation of $r = 0.42$. Which assessment is correct?

- A) The values are consistent because slope and correlation use different units.
- B) The values are consistent whenever the intercept is negative.
- C) The correlation should equal the reciprocal of the slope.
- D) The values cannot both be correct because a least-squares slope and correlation must have the same sign.

Questions APSTAT-MCQ12-012 to APSTAT-MCQ12-014 refer to the following stimulus.

For a randomly selected service visit, let X be the number of follow-up calls required. The probability distribution is shown below.

X	0	1	2	3
$P(X)$	0.10	0.25	0.40	0.25

APSTAT-MCQ12-012

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For X with $P(X = 0) = 0.10$, $P(X = 1) = 0.25$, $P(X = 2) = 0.40$, and $P(X = 3) = 0.25$, what is $P(X = 3 \mid X \geq 1)$?

- A) $0.25/0.90 \approx 0.278$
- B) $0.25/0.65 \approx 0.385$
- C) 0.25
- D) 0.90

APSTAT-MCQ12-013

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For the follow-up-call distribution $P(X = 0, 1, 2, 3) = (0.10, 0.25, 0.40, 0.25)$, what is the expected number of follow-up calls per service visit?

- A) 1.50
- B) 1.80
- C) 2.00

D) 2.15

APSTAT-MCQ12-014

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For the distribution $P(X = 0, 1, 2, 3) = (0.10, 0.25, 0.40, 0.25)$, which value is a median of X , and why?

- A) 1, because $P(X \leq 1) = 0.35$
- B) 1.8, because every median equals the expected value
- C) 2, because $P(X \leq 2) = 0.75$ and $P(X \geq 2) = 0.65$
- D) 3, because 3 is the largest possible value

APSTAT-MCQ12-015

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a random sample of 120 accounts, 78 have enabled a security feature. For $H_0: p = 0.60$, what is the one-proportion z statistic?

- A) Approximately 1.12
- B) Approximately 0.65
- C) Approximately 1.77
- D) Approximately 2.24

APSTAT-MCQ12-016

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A travel survey asks respondents who used several transportation modes to report one "usual mode," but gives no definition of usual. Which revision best reduces ambiguity while still producing one categorical response per person?

- A) Ask respondents to choose whichever mode they like most.
- B) Let interviewers decide which reported mode appears usual.

- C) Keep the original wording and remove people who name more than one mode.
- D) Define a reference trip and rule, such as the mode covering the greatest distance on the most recent weekday commute, and provide exhaustive response categories.

APSTAT-MCQ12-017

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Five paired differences, defined as after – before, are –4, –1, 0, 2, and –2. What is the t statistic for $H_0: \mu_d = 0$?

- A) –2.24
- B) –1.00
- C) 0
- D) 1.00

APSTAT-MCQ12-018

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

For the same data and two-sided method, an analyst reports a 95% confidence interval for $p_1 - p_2$ of (–0.02, 0.11) and a p-value of 0.010 for testing $H_0: p_1 - p_2 = 0$. Which assessment is correct?

- A) Both reports are consistent because 0.010 is inside the confidence interval.
- B) Both reports are consistent because the interval contains positive values.
- C) The reports prove that p_1 and p_2 are equal.
- D) The reports are inconsistent: an interval containing 0 corresponds to failure to reject the zero difference at the 5% level, not to $p = 0.010$.

APSTAT-MCQ12-019

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Nine ordered observations all lie between 10 and 30. Their mean and median are both 20. Two additional observations, 5 and 35, are added. Which statement is guaranteed?

- A) The mean becomes 22.5, while the median stays 20.
- B) The mean stays 20, while the median becomes 22.5.
- C) Both the mean and the median remain 20.
- D) The mean remains 20, but the new median cannot be determined.

APSTAT-MCQ12-020

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Two different serial-number digits are sampled without replacement from 0 through 9. What is the probability that their product is even?

- A) $\frac{7}{9}$
- B) $\frac{5}{9}$
- C) $\frac{1}{2}$
- D) $\frac{2}{9}$

APSTAT-MCQ12-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A violin technician wants to study tuning drift during rehearsals. Which question is suited to least-squares regression using rehearsal-level data?

- A) What proportion of rehearsals have high humidity?
- B) How well does measured room humidity predict the quantitative change in tuning frequency during the same rehearsal?
- C) Is the brand of violin associated with the categorical type of music performed?
- D) What is the average humidity across all rehearsal halls?

APSTAT-MCQ12-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

An automated screen classifies packages as damaged or not damaged. Investigators want population estimates of the screen's false-positive and false-negative proportions, but damaged packages are rare. Which validation plan is best?

- A) Randomly sample packages within both screen-positive and screen-negative strata, apply a blinded gold-standard inspection, and retain the sampling fractions for weighted analysis.
- B) Inspect only packages labeled damaged and treat every screen-positive package as a true positive.
- C) Ask screen operators to submit examples on which they believe the algorithm is wrong.
- D) Validate the same convenient batch repeatedly until both error counts exceed 10.

APSTAT-MCQ12-023

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A display places four nominal music genres in alphabetical order and connects their audience proportions with line segments. Which critique is most appropriate?

- A) Proportions cannot be displayed graphically.
- B) Alphabetical order proves the proportions follow a time trend.
- C) Connecting nominal categories suggests a continuous progression that the categories do not possess; separate bars would communicate the comparison more clearly.
- D) The line segments make the numerical proportions incorrect.

APSTAT-MCQ12-024

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

A researcher is planning a confidence interval for a population mean and wants to choose the sample size before collecting data. Which information is needed for the usual planning calculation?

- A) The sample mean and the final p-value
- B) The population mean and the observed t statistic
- C) Only the population size, regardless of precision
- D) The desired confidence level, target margin of error, and a defensible planning value for the population standard deviation

APSTAT-MCQ12-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Three detectors operate independently. Their trigger probabilities on a routine scan are 0.20, 0.30, and 0.40. What is the probability that at least two detectors trigger?

- A) 0.024
- B) 0.176
- C) 0.188
- D) 0.212

APSTAT-MCQ12-026

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A new process is declared noninferior if $p_{\text{new}} - p_{\text{standard}} > -0.05$. A 95% confidence interval for this difference is $(-0.03, 0.06)$. Which conclusion is supported?

- A) The new process is superior because every plausible difference is positive.
- B) The interval supports noninferiority because its lower endpoint exceeds -0.05 , but it does not establish superiority because 0 is included.
- C) The processes have been proved identical because the interval contains 0.
- D) Noninferiority fails because the interval includes some negative values.

APSTAT-MCQ12-027

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A nutrition study wants more accurate information about when participants snack during the day. Which protocol best reduces end-of-week recall error?

- A) Send prompts at randomly selected times and have participants record current or just-completed eating behavior immediately using a standardized form.

- B) Ask participants at week's end to remember every snack and its exact time.
- C) Ask each participant to report only the day that seemed most typical.
- D) Let interviewers fill gaps with the snack pattern they expect for that participant.

APSTAT-MCQ12-028

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For the same estimate of a population mean and standard error $SE = 1.5$, a 90% t-interval uses $t^* = 1.729$ and a 98% t-interval uses $t^* = 2.528$. By approximately how much is the full width of the 98% interval greater than the full width of the 90% interval?

- A) 0.80
- B) 1.20
- C) 2.40
- D) 4.80

APSTAT-MCQ12-029

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

An agronomist compares three soil amendments on 27 plots. Soil texture is an important source of variability. Which design uses blocking effectively?

- A) Apply amendment 1 to the nine sandiest plots, amendment 2 to the next nine, and amendment 3 to the nine least sandy plots.
- B) Let each plot manager select an amendment.
- C) Randomly assign nine plots to each amendment without recording soil texture.
- D) Form nine blocks of three plots with similar soil texture, then randomly assign the three amendments so each appears once within every block.

APSTAT-MCQ12-030

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a randomization test for a difference in proportions, 1,000 shuffled differences are generated under H_0 . Of these, 18 are at least as large as the observed positive difference and 22 are at least as far in the negative direction. What is the approximate two-sided p-value?

- A) 0.018
- B) 0.022
- C) 0.040
- D) 0.080

APSTAT-MCQ12-031

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

The interquartile range of daily charging times is 14 minutes. Which interpretation is correct?

- A) Every charging time is within 14 minutes of the median.
- B) The middle 50% of charging times span 14 minutes from Q_1 to Q_3 .
- C) The mean charging time is 14 minutes greater than the median.
- D) Exactly 14% of charging times are outside the quartiles.

APSTAT-MCQ12-032

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

An analyst wants to examine whether subscription type is associated with renewal status. Which comparison directly addresses the question?

- A) Compare $P(\text{renewal} \mid \text{premium})$ with $P(\text{renewal} \mid \text{basic})$.
- B) Compare $P(\text{premium})$ with $P(\text{basic})$.
- C) Compare the number of renewed subscriptions with the total number of subscriptions, ignoring type.
- D) Compare the mean numerical codes assigned to the two subscription types.

Questions APSTAT-MCQ12-033 to APSTAT-MCQ12-035 refer to the following stimulus.

For paired quantitative data, the least-squares line predicting Y from X is $\hat{y} = 20 + 3x$ and the correlation is $r = 0.70$. A reflected response is defined by $W = 100 - Y$.

APSTAT-MCQ12-033

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Under the response reflection $W = 100 - Y$, how are the fitted intercept and slope 20 and 3 transformed, respectively?

- A) 120 and 3
- B) 100 and -3
- C) 80 and -3
- D) 80 and 3

APSTAT-MCQ12-034

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Replacing every response y by $100 - y$ has what effect on an original correlation of $r = 0.70$?

- A) Its magnitude is squared to 0.49, with positive sign.
- B) Its magnitude remains 0.70, and its sign becomes negative.
- C) Neither its magnitude nor its sign changes.
- D) Its magnitude becomes 1 because reflection is exact.

APSTAT-MCQ12-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

For one observation, the original residual is $y - \hat{y} = 5$. Which statement describes its residual after the response is reflected to $W = 100 - Y$?

- A) The new residual is 105 because the response was reflected around 100.
- B) The new residual remains 5 because correlation magnitude is unchanged.
- C) The new residual is -5 , so its contribution to SSE changes from 25 to -25 .
- D) The new residual is -5 , and its squared contribution to SSE remains 25.

APSTAT-MCQ12-036

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

The estimated standard error for a difference of two independent sample means is 4.00 units. If both sample sizes are doubled while both sample standard deviations remain unchanged, what is the new standard error?

- A) $4/\sqrt{2} \approx 2.83$ units
- B) 2.00 units
- C) $4\sqrt{2} \approx 5.66$ units
- D) 8.00 units

APSTAT-MCQ12-037

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

An agricultural organization wants to test whether more than half of all regenerative farms use automated soil monitoring. Which hypotheses are appropriate?

- A) $H_0: \hat{p} = 0.50$; $H_a: \hat{p} > 0.50$
- B) $H_0: p = 0.50$; $H_a: p > 0.50$
- C) $H_0: p > 0.50$; $H_a: p = 0.50$
- D) $H_0: p = 0.50$; $H_a: p \neq 0.50$

APSTAT-MCQ12-038

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A study seeks accurate weekly screen-time measurements from a random sample of consenting smartphone users. Which method best reduces recall and social-desirability error?

- A) Ask users to estimate last month's average from memory during a public interview.
- B) Ask friends of each user to estimate the user's screen time.
- C) Let users report only days they consider typical.

D) With informed consent, collect standardized passive device logs for the prespecified week and verify that the same definition of screen time is used on every device.

APSTAT-MCQ12-039

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A committee of 3 people is selected at random from 5 architects and 4 engineers. What is the probability that the committee includes at least one person from each profession?

- A) $\frac{5}{6}$
- B) $\frac{5}{14}$
- C) $\frac{1}{2}$
- D) $\frac{11}{14}$

APSTAT-MCQ12-040

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

Using the same sample estimate and standard error, a report gives a 90% confidence interval (0.12, 0.28), a 95% interval (0.10, 0.30), and a 99% interval (0.14, 0.26). What is wrong with the report?

- A) All three intervals must have different midpoints.
- B) A 90% interval must be wider than a 95% interval.
- C) The 99% interval should be the widest, but the reported 99% interval is the narrowest.
- D) No confidence interval may contain the sample estimate.

APSTAT-MCQ12-041

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A university wants an approximately self-weighting sample of students, but course sections vary greatly in enrollment. Which two-stage plan best prevents students in small sections from being overrepresented?

- A) Select sections with probability proportional to enrollment, then randomly select the same number of students within each chosen section.
- B) Select sections with equal probability, then randomly select the same number of students within each chosen section.
- C) Choose the smallest sections and survey every student in them.
- D) Invite instructors to nominate students who regularly attend class.

APSTAT-MCQ12-042

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Two algorithms measure each of 16 audio recordings. For the paired differences $d = \text{algorithm A} - \text{algorithm B}$, $\bar{x}_d = -2.5$ seconds and $s_d = 4.0$ seconds. Using $t^* = 2.131$, what is a 95% confidence interval for μ_d ?

- A) $(-10.50, 5.50)$ seconds
- B) $(-4.631, -0.369)$ seconds
- C) $(-3.033, -1.967)$ seconds
- D) $(-2.631, -2.369)$ seconds

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ12-001, Correct: B, Correct Answer: Minimum 20, Q1 = 32, median 40, Q3 = 46, maximum 52, Explanation: Because $Y = 60 - 2X$ is decreasing, the order reverses. Transforming the original maximum, Q3, median, Q1, and minimum gives 20, 32, 40, 46, and 52, respectively.

APSTAT-MCQ12-002, Correct: D, Correct Answer: The text-reminder population proportion is estimated to be 4 to 12 percentage points lower than the email-reminder population proportion., Explanation: Every plausible value for $p_{\text{text}} - p_{\text{email}}$ is negative. The endpoints -0.12 and -0.04 correspond to a text response proportion between 12 and 4 percentage points lower than the email response proportion.

APSTAT-MCQ12-003, Correct: A, Correct Answer: $1/10$, Explanation: $P(\text{both amber}) = (3/12)(2/11) = 1/22$. Also, $P(\text{at least one amber}) = 1 - (9/12)(8/11) = 5/11$. Therefore $P(\text{both amber} \mid \text{at least one amber}) = (1/22)/(5/11) = 1/10$.

APSTAT-MCQ12-004, Correct: C, Correct Answer: $H_0: \mu_d = 0$; $H_a: \mu_d < 0$, Explanation: With d defined as automated minus manual time, a reduction from automation corresponds to negative differences. The hypotheses must use the population mean paired difference μ_d , not the sample mean or independent-sample parameters.

APSTAT-MCQ12-005, Correct: B, Correct Answer: 26, Explanation: A least-squares line passes through $(\bar{x}, \square)$. Substituting $\bar{x} = 5$ gives $\square = 18 + 1.6(5) = 26$.

APSTAT-MCQ12-006, Correct: A, Correct Answer: Construct a list of individual apartment units in the study area and randomly select units from that list., Explanation: The target population consists of apartments, so the sampling frame should give every apartment unit a selection opportunity. A building-level list with one row per building underrepresents apartments in large buildings unless a multistage design and appropriate weights are used.

APSTAT-MCQ12-007, Correct: D, Correct Answer: The usual large-count condition is not met because sample B has only 7 failures., Explanation: A two-proportion z-interval ordinarily requires at least 10 successes and 10 failures in each group. Sample B has only 7 failures, so the condition is not met even though the total sample size exceeds 30.

APSTAT-MCQ12-008, Correct: C, Correct Answer: Values of $\hat{p}$ typically vary by about 0.0485 from the population proportion 0.38 across repeated samples., Explanation: The standard deviation of a sampling distribution measures sample-to-sample variability of the statistic around its mean. It does not describe a changing population parameter or guarantee a fixed range for every $\hat{p}$.

APSTAT-MCQ12-009, Correct: C, Correct Answer: Approximately 2.11, Explanation: The corrected sum of squares is $\sum(x - \bar{x})^2 = \sum x^2 - (\sum x)^2/n = 530 - 70^2/10 = 40$. Thus $s = \sqrt{40/(10 - 1)} = \sqrt{(40/9)} \approx 2.11$.

APSTAT-MCQ12-010, Correct: B, Correct Answer: 2.60, Explanation: The statistic is $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$. Increasing n from 25 to 100 halves the standard error while the numerator and s stay fixed, so t doubles from 1.30 to 2.60.

APSTAT-MCQ12-011, Correct: D, Correct Answer: The values cannot both be correct because a least-squares slope and correlation must have the same sign., Explanation: The slope satisfies $b_1 = r(s_y/s_x)$. Because both standard deviations are positive, b_1 and r must share a sign. A negative slope cannot accompany a positive correlation for the same paired data.

APSTAT-MCQ12-012, Correct: A, Correct Answer: $0.25/0.90 \approx 0.278$, Explanation: The conditioning event $X \geq 1$ has probability $1 - 0.10 = 0.90$. Because $X = 3$ is contained in that event, the conditional probability is $0.25/0.90 \approx 0.278$.

APSTAT-MCQ12-013, Correct: B, Correct Answer: 1.80, Explanation: $E(X) = 0(0.10) + 1(0.25) + 2(0.40) + 3(0.25) = 0.25 + 0.80 + 0.75 = 1.80$ follow-up calls.

APSTAT-MCQ12-014, Correct: C, Correct Answer: 2, because $P(X \leq 2) = 0.75$ and $P(X \geq 2) = 0.65$, Explanation: A median has at least half the probability at or below it and at least half at or above it. Both conditions hold for 2; the cumulative probability through 1 is only 0.35, so 1 is not a median.

APSTAT-MCQ12-015, Correct: A, Correct Answer: Approximately 1.12, Explanation: Here $\hat{p} = 78/120 = 0.65$. Under H_0 , $SE\hat{p} = \sqrt{0.60(0.40)/120} = \sqrt{0.002} \approx 0.04472$, so $z = (0.65 - 0.60)/0.04472 \approx 1.12$.

APSTAT-MCQ12-016, Correct: D, Correct Answer: Define a reference trip and rule, such as the mode covering the greatest distance on the most recent weekday commute, and provide exhaustive response categories., Explanation: A fixed reference trip and operational rule give "usual" the same meaning for every respondent. Exhaustive categories allow one coded response without relying on personal preferences, interviewer judgment, or deletion of multimode travelers.

APSTAT-MCQ12-017, Correct: B, Correct Answer: -1.00 , Explanation: The mean difference is $\bar{x}_d = -1$. The squared deviations from -1 sum to 20, so $s_d = \sqrt{(20/4)} = \sqrt{5}$ and $SE = \sqrt{5}/\sqrt{5} = 1$. Thus $t = (-1 - 0)/1 = -1.00$.

APSTAT-MCQ12-018, Correct: D, Correct Answer: The reports are inconsistent: an interval containing 0 corresponds to failure to reject the zero difference at the 5% level, not to $p = 0.010$., Explanation: For matching two-sided methods, a 95% interval contains the null value exactly when the corresponding test does not reject at $\alpha = 0.05$. A reported two-sided p-value of 0.010 would instead require the 95% interval to exclude 0.

APSTAT-MCQ12-019, Correct: C, Correct Answer: Both the mean and the median remain 20., Explanation: The original total is $9(20) = 180$; adding 5 and 35 raises the total to 220 over 11 values, so the new mean is 20. The additions place one value below and one above all original observations, leaving the original fifth value, 20, in the new sixth and middle position.

APSTAT-MCQ12-020, Correct: A, Correct Answer: $7/9$, Explanation: The product is odd only if both selected digits are odd. Of $C(10, 2) = 45$ unordered pairs, $C(5, 2) = 10$ contain two odd digits, so $P(\text{even product}) = 1 - 10/45 = 35/45 = 7/9$.

APSTAT-MCQ12-021, Correct: B, Correct Answer: How well does measured room humidity predict the quantitative change in tuning frequency during the same rehearsal?, Explanation: Least-squares regression requires paired quantitative measurements on the same observational units. Rehearsal-level humidity and tuning-frequency change meet that requirement and align the explanatory and response roles with the prediction goal.

APSTAT-MCQ12-022, Correct: A, Correct Answer: Randomly sample packages within both screen-positive and screen-negative strata, apply a blinded gold-standard inspection, and retain the sampling fractions for weighted analysis., Explanation: Gold-standard review in both algorithm strata permits estimation of false positives and false negatives. Random selection and retained sampling fractions support valid weighted population estimates while allowing efficient oversampling of the rare screen-positive stratum.

APSTAT-MCQ12-023, Correct: C, Correct Answer: Connecting nominal categories suggests a continuous progression that the categories do not possess; separate bars would communicate the comparison more clearly., Explanation: Nominal genres have no inherent numerical spacing or continuous order. A connected line invites an unsupported trend interpretation, whereas separate bars emphasize comparison among distinct categories without implying continuity.

APSTAT-MCQ12-024, Correct: D, Correct Answer: The desired confidence level, target margin of error, and a defensible planning value for the population standard deviation, Explanation: Mean-interval planning uses $n \approx (\text{critical value} \times \text{planning } \sigma / \text{target margin})^2$, with upward rounding. The unknown population mean and a future p-value are not inputs to the precision calculation.

APSTAT-MCQ12-025, Correct: D, Correct Answer: 0.212, Explanation: Exactly two triggers have probability $(0.20)(0.30)(0.60) + (0.20)(0.70)(0.40) + (0.80)(0.30)(0.40) = 0.188$. All three trigger with probability $(0.20)(0.30)(0.40) = 0.024$, so $P(\text{at least two}) = 0.188 + 0.024 = 0.212$.

APSTAT-MCQ12-026, Correct: B, Correct Answer: The interval supports noninferiority because its lower endpoint exceeds -0.05 , but it does not establish superiority because 0 is included., Explanation:

Every value in the interval is above the noninferiority boundary -0.05 , ruling out losses larger than the allowed margin at the stated confidence level. Because the interval crosses 0, the data do not establish a positive advantage for the new process.

APSTAT-MCQ12-027, Correct: A, Correct Answer: Send prompts at randomly selected times and have participants record current or just-completed eating behavior immediately using a standardized form., Explanation: Random momentary prompts spread measurements across the day and shorten the recall window. End-of-week reconstruction, subjective typical-day selection, and interviewer-filled gaps can create systematic recall or expectation error.

APSTAT-MCQ12-028, Correct: C, Correct Answer: 2.40, Explanation: A two-sided interval has full width $2(t^*)(SE)$. The increase is $2(2.528 - 1.729)(1.5) = 2.397$, which is approximately 2.40 units.

APSTAT-MCQ12-029, Correct: D, Correct Answer: Form nine blocks of three plots with similar soil texture, then randomly assign the three amendments so each appears once within every block., Explanation: Grouping similar plots controls the known soil-texture source of variation. Random assignment of all three treatments within every block keeps treatment from being confounded with texture and permits comparisons within matched triples.

APSTAT-MCQ12-030, Correct: C, Correct Answer: 0.040, Explanation: A two-sided randomization p-value counts simulated differences at least as extreme in both directions. The approximate p-value is $(18 + 22)/1,000 = 40/1,000 = 0.040$.

APSTAT-MCQ12-031, Correct: B, Correct Answer: The middle 50% of charging times span 14 minutes from Q1 to Q3., Explanation: $IQR = Q3 - Q1$ measures the width of the middle half of a distribution. It does not constrain every observation or describe a difference between the mean and median.

APSTAT-MCQ12-032, Correct: A, Correct Answer: Compare $P(\text{renewal} \mid \text{premium})$ with $P(\text{renewal} \mid \text{basic})$., Explanation: Association is assessed by comparing the conditional renewal distributions across subscription categories. Marginal category frequencies or arbitrary numerical codes do not measure how renewal varies by type.

APSTAT-MCQ12-033, Correct: C, Correct Answer: 80 and -3 , Explanation: Since $W = 100 - Y$, every fitted response is reflected in the same way: $\hat{w} = 100 - \hat{y} = 100 - (20 + 3x) = 80 - 3x$. Thus the new intercept is 80 and the new slope is -3 .

APSTAT-MCQ12-034, Correct: B, Correct Answer: Its magnitude remains 0.70, and its sign becomes negative., Explanation: Subtracting Y from a constant multiplies its standardized values by -1 . Therefore the direction of association reverses while its strength is unchanged, giving $r_{XW} = -r_{XY} = -0.70$.

APSTAT-MCQ12-035, Correct: D, Correct Answer: The new residual is -5 , and its squared contribution to SSE remains 25., Explanation: The transformed residual is $w - \hat{w} = (100 - y) - (100 - \hat{y}) = -(y - \hat{y}) = -5$. Reflection changes the residual's sign but not its square, so the observation still contributes 25 to SSE.

APSTAT-MCQ12-036, Correct: A, Correct Answer: $4/\sqrt{2} \approx 2.83$ units, Explanation: Each variance contribution s^2/n is halved when its sample size doubles. Their sum is therefore halved, so the standard error is multiplied by $\sqrt{(1/2)}$: $4/\sqrt{2} \approx 2.83$ units.

APSTAT-MCQ12-037, Correct: B, Correct Answer: $H_0: p = 0.50$; $H_a: p > 0.50$, Explanation: The hypotheses concern the population proportion p . The null uses the boundary value 0.50, and “more than half” determines the right-tailed alternative $p > 0.50$.

APSTAT-MCQ12-038, Correct: D, Correct Answer: With informed consent, collect standardized passive device logs for the prespecified week and verify that the same definition of screen time is used on every device., Explanation: Standardized logs record behavior directly and consistently, reducing recall and social-desirability error. Selective “typical” days or estimates from users and friends introduce subjective measurement error.

APSTAT-MCQ12-039, Correct: A, Correct Answer: $5/6$, Explanation: There are $C(9, 3) = 84$ possible committees. The 10 all-architect and 4 all-engineer committees fail the requirement, so the desired probability is $1 - (10 + 4)/84 = 70/84 = 5/6$.

APSTAT-MCQ12-040, Correct: C, Correct Answer: The 99% interval should be the widest, but the reported 99% interval is the narrowest., Explanation: With the estimate and standard error fixed, higher confidence requires a larger critical value and margin of error. All intervals should share midpoint 0.20, with the 99% interval wider than the 95% and 90% intervals.

APSTAT-MCQ12-041, Correct: A, Correct Answer: Select sections with probability proportional to enrollment, then randomly select the same number of students within each chosen section., Explanation: A student's section has a selection chance proportional to its size, followed by an inverse-size within-section chance when the same number is sampled. Those factors offset, giving students approximately equal overall inclusion probabilities.

APSTAT-MCQ12-042, Correct: B, Correct Answer: $(-4.631, -0.369)$ seconds, Explanation: The paired standard error is $s_d/\sqrt{n} = 4/\sqrt{16} = 1$ second. The interval is $-2.5 \pm 2.131(1)$, which equals $(-4.631, -0.369)$ seconds.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ12

· APSTAT-MCQ12-001 to 005: Unit 1 applies the decreasing transformation $Y = 60 - 2X$ to a five-number summary and reverses the quartiles; Unit 3 interprets a fully negative text-minus-email reminder-response interval; Unit 2 conditions two without-replacement tag draws on at least one amber tag; Unit 4 formulates a left-tailed paired-mean hypothesis for automated-minus-manual processing time; Unit 5 uses the least-squares mean-point property to recover $\square$ from $\bar{x}$ and $\hat{y} = 18 + 1.6x$. Misconceptions include failing to reverse quantile order under a decreasing transformation, reversing a difference-of-proportions direction, using an unconditional denominator, putting $\bar{x}_d$ in hypotheses, and treating the regression line as unrelated to the sample means.

· APSTAT-MCQ12-006 to 010: Unit 1 replaces a building-level address frame with an apartment-unit frame so large buildings are not underrepresented; Unit 3 finds the two-proportion large-count failure from seven nonsuccesses in one group; Unit 2 interprets 0.0485 as repeated-sample variability of $\hat{p}$ around 0.38; Unit 1 uses $n = 10$, $\Sigma x = 70$, and $\Sigma x^2 = 530$ to obtain $s = \sqrt{(40/9)} \approx 2.11$; Unit 4 quadrupling n from 25 to 100 with fixed $\bar{x}$ and s doubles a one-mean t statistic from 1.30 to 2.60. Misconceptions include treating one building row as one apartment, using total n for large counts, treating a parameter as variable, omitting the correction $(\Sigma x)^2/n$ or $n - 1$ divisor, and assuming t is unchanged when its standard error halves.

· APSTAT-MCQ12-011 to 015: Unit 5 rejects an internally inconsistent regression report with negative slope -1.8 but positive $r = 0.42$ because slope and correlation signs must agree; Unit 2 a service-visit

distribution on 0–3 follow-up calls calculates $P(X = 3 \mid X \geq 1) = 0.25/0.90$, $E(X) = 1.80$, and median 2 from both cumulative sides; Unit 3 uses 78 of 120 against $p_0 = 0.60$ to calculate $z \approx 1.12$.

Misconceptions include allowing slope units or intercept to reverse the sign relationship, conditioning on the wrong probability mass, choosing the most likely count as the mean, equating mean and median, and using an observed rather than null standard error.

· APSTAT-MCQ12-016 to 020: Unit 1 operationalizes one “usual transportation mode” through a fixed reference commute, greatest-distance rule, and exhaustive categories; Unit 4 paired differences $-4, -1, 0, 2, -2$ give $\bar{x}_d = -1$, $SE = 1$, and $t = -1$; Unit 3 detects incompatibility between a 95% interval containing 0 and a matching two-sided p-value 0.010; Unit 1 adding symmetric exterior observations 5 and 35 to nine values with mean and median 20 preserves both centers; Unit 2 two serial-number digits use the complement of 10 all-odd pairs among 45 to obtain even-product probability $7/9$.

Misconceptions include leaving ambiguous survey terminology undefined, ignoring the $n - 1$ variance divisor, treating a p-value as an interval member, overlooking the unchanged total and middle rank, and counting only even-even pairs for an even product.

· APSTAT-MCQ12-021 to 025: Unit 5 pairs rehearsal humidity with same-rehearsal tuning-frequency drift for a regression question; Unit 3 stratifies algorithm-positive and algorithm-negative packages for blinded gold-standard validation with retained sampling fractions; Unit 1 critiques line segments across alphabetically ordered nominal music genres; Unit 4 mean-interval sample-size planning requires confidence, target margin, and a defensible σ value; Unit 2 three independent detector probabilities 0.20, 0.30, and 0.40 give $P(\text{at least two triggers}) = 0.212$ by summing three exactly-two patterns and the all-three pattern. Misconceptions include unpaired or categorical regression variables, validating only screen positives, implying continuity among nominal categories, using a future sample mean or p-value in planning, and counting only one trigger pattern.

· APSTAT-MCQ12-026 to 030: Unit 3 a new-minus-standard interval $(-0.03, 0.06)$ clears a -0.05 noninferiority margin without establishing superiority; Unit 1 random momentary prompts record snack behavior immediately to reduce weekly recall error; Unit 4 with $SE = 1.5$ compares $t^* = 1.729$ and 2.528 to find that a 98% interval is about 2.40 units wider than a 90% interval; Unit 1 soil-texture matched triples receive all three randomly assigned amendments within every block; Unit 3 a two-sided randomization p-value adds 18 positive-tail and 22 negative-tail shuffled results to obtain 0.040. Misconceptions include requiring every plausible effect to be positive for noninferiority, relying on end-week reconstruction, comparing margins instead of full widths, assigning amendments by soil rank rather than within blocks, and counting only the observed-direction tail.

· APSTAT-MCQ12-031 to 035: Unit 1 interprets a 14-minute IQR as the Q1-to-Q3 span of the middle half; Unit 2 formulates an association question by comparing renewal conditional on premium versus basic subscription; Unit 5 reflects response Y to $W = 100 - Y$, transforming $\hat{y} = 20 + 3x$ and $r = 0.70$ into $\hat{w} = 80 - 3x$ and $r_{XW} = -0.70$ while reversing a residual's sign but preserving its squared SSE contribution. Misconceptions include placing all observations within one IQR of the median, comparing marginal rather than conditional probabilities, failing to transform the regression intercept and slope together, retaining the correlation sign, and treating a negative residual as a negative SSE contribution.

· APSTAT-MCQ12-036 to 040: Unit 4 doubling both independent-sample sizes changes SE 4 to $4/\sqrt{2}$; Unit 3 tests a regenerative-farm automated-monitoring proportion against 0.50 with a right-tailed population parameter; Unit 1 standardized passive smartphone logs reduce screen-time recall and social-desirability error; Unit 2 a three-person committee from five architects and four engineers uses a complement to obtain mixed-profession probability $5/6$; Unit 3 identifies an impossible confidence-level ordering because a reported 99% interval is narrower than 90% and 95% intervals from the same estimate and SE . Misconceptions include halving SE when n merely doubles, using $\hat{p}$ in hypotheses, relying on self-selected typical days, omitting all-one-profession committees, and narrowing an interval as confidence rises.

- APSTAT-MCQ12-041 to 042: Unit 1 a two-stage student sample selects course sections with probability proportional to enrollment and then the same number of students per selected section for approximately equal inclusion probabilities; Unit 4 paired algorithm timing differences with $n = 16$, $\bar{x}_d = -2.5$, $s_d = 4$, and $t^* = 2.131$ yield interval $(-4.631, -0.369)$ seconds. Misconceptions include selecting unequal-size sections with equal probability and omitting $\sqrt{n}$ from the paired standard error.
- Unit counts for APSTAT-MCQ12: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: service-visit follow-up-call probability distribution, APSTAT-MCQ12-012 to 014; reciprocal-role regression summaries, APSTAT-MCQ12-033 to 035.